

GCC

Tags:

Time Limit: 1 s Memory Limit: 64 MB
Submission: 394 AC: 89 Score: 98.25

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Codes

AI Code Tips

Description

The GNU Compiler Collection (usually shortened to GCC) is a compiler system produced by the GNU Project supporting various programming languages. But it doesn't contains the math operator "!".

In mathematics the symbol represents the factorial operation. The expression $n!$ means "the product of the integers from 1 to n ". For example, $4!$ (read four factorial) is $4 \times 3 \times 2 \times 1 = 24$. ($0!$ is defined as 1, which is a neutral element in multiplication, not multiplied by anything.)

We want you to help us with this formation: $(0! + 1! + 2! + 3! + 4! + \dots + n!) \% m$

Input

The first line consists of an integer T , indicating the number of test cases.

Each test on a single consists of two integer n and m .

Output

Output the answer of $(0! + 1! + 2! + 3! + 4! + \dots + n!) \% m$.

Constrains

$0 < T \leq 20$

$0 \leq n < 10^{100}$ (without leading zero)

$0 < m < 1000000$

Samples

input	Copy
1 10 861017	
output	Copy
593846	

Source

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 14:58:08